

Brain Myelin Water Fraction and Diffusion Tensor Imaging Atlases for 9-10 Year-Old Children

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ABSTRACT

BACKGROUND AND PURPOSE: Myelin water imaging (MWI) and diffusion tensor imaging (DTI) provide information about myelin and axon-related brain microstructure, which can be useful for investigating normal brain development and many childhood brain disorders. While pediatric DTI atlases exist, there are no pediatric MWI atlases available for the 9-10 years old age group. As myelination and structural development occurs throughout childhood and adolescence, studies of pediatric brain pathologies must use age-specific MWI and DTI healthy control data. We created atlases of myelin water fraction (MWF) and DTI metrics for healthy children aged 9-10 years for use as normative data in pediatric neuroimaging studies.

METHODS: 3D-T₁, DTI, and MWI scans were acquired from 20 healthy children (mean age: 9.6 years, range: 9.2-10.3 years, 4 females). ANTs and FSL registration were used to create quantitative MWF and DTI atlases. Region of interest (ROI) analysis in nine white matter regions was used to compare pediatric MWF with adult MWF values from a recent study and to investigate the correlation between pediatric MWF and DTI metrics.

RESULTS: Adults had significantly higher MWF than the pediatric cohort in seven of the nine white matter ROIs, but not in the genu of the corpus callosum or the cingulum. In the pediatric data, MWF correlated significantly with mean diffusivity, but not with axial diffusivity, radial diffusivity, or fractional anisotropy.

CONCLUSIONS: Normative MWF and DTI metrics from a group of 9-10 year old healthy children provide a resource for comparison to pathologies. The age-specific atlases are ready for use in pediatric neuroimaging research and can be accessed: <https://sourceforge.net/projects/pediatric-mri-myelin-diffusion/>.

Keywords: Atlas, diffusion tensor imaging, myelin water fraction, myelin water imaging, pediatric.

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Introduction

Quantifying pediatric brain myelination is important to understand both healthy development and the many pediatric brain diseases that affect white matter microstructure. While all major white matter tracts in the brain are present by the end of gestation,¹ myelination of these tracts is a process which continues throughout infancy, childhood, and adolescence.² This myelination allows for more rapid and efficient neural com-

munication and has been linked to many aspects of cognitive development.³⁻⁶ While this myelination is occurring, the volume of white matter in the brain also increases and changes to axonal properties can be seen in diffusion imaging studies.⁷⁻⁹

Age-specific quantitative MRI control data are needed when investigating pediatric brain pathologies due to the rapid alterations occurring throughout the brain during development. An age-specific structural template is also necessary, as registration

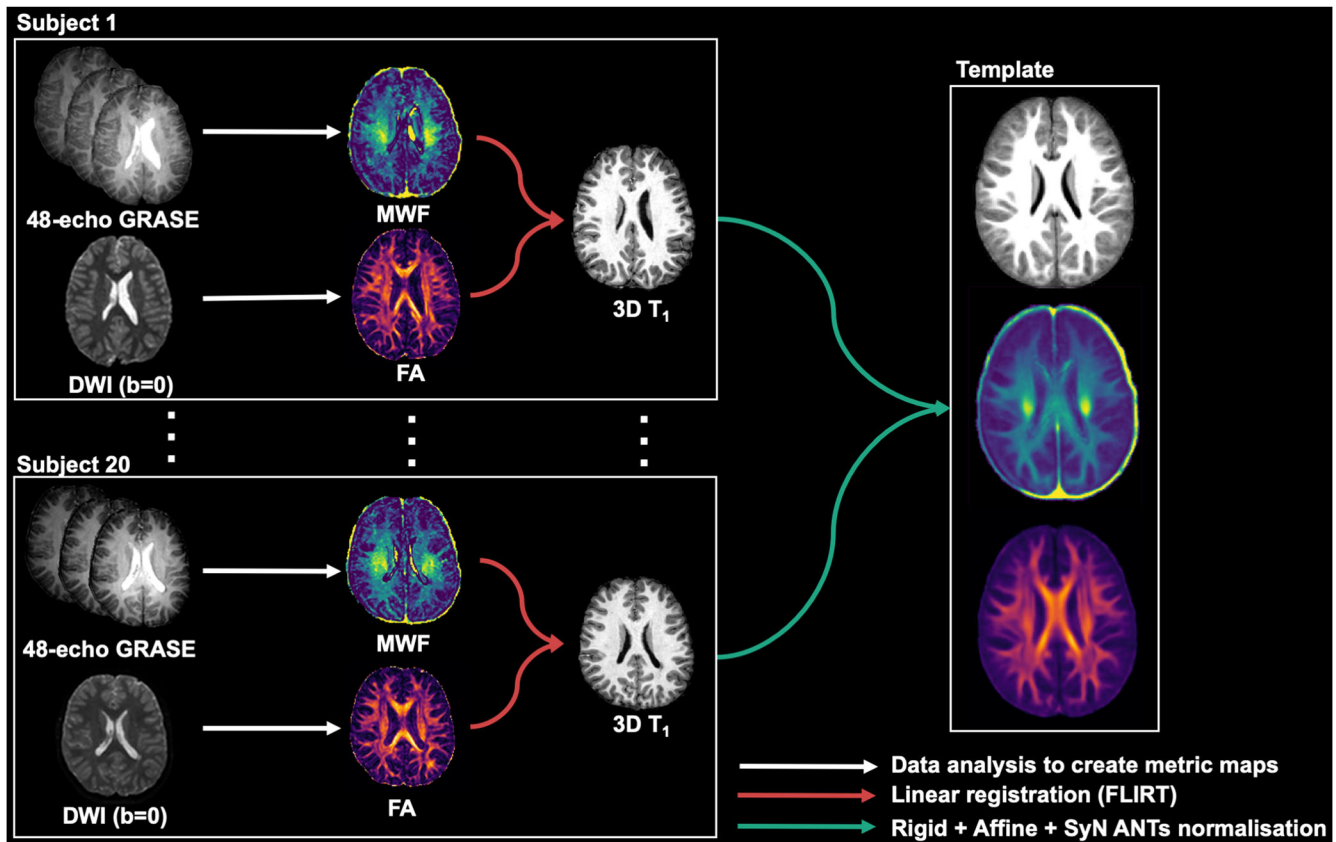


Fig 1. Template creation process: metric maps were created from Diffusion-Weighted Imaging (DWI) and 48-echo Gradient and Spin Echo (GRASE) data. Myelin Water Fraction (MWF) and Fractional Anisotropy (FA) maps were registered to each participant's 3D-T₁ scan via an FSL affine transformation. Mean diffusivity, radial diffusivity, and axial diffusivity maps were registered using the same affine registration as FA. The 3D-T₁ scans from each participant were co-registered to create the template space using rigid, affine, and symmetric diffeomorphic normalization transformation algorithms in Advanced Normalization Tools (ANTs). The warps produced were used to transform the metric data into the template space where the metric maps were averaged to create atlases.

of pediatric brain data to an adult standard space can cause increased deformation and bias.^{10,11} Recruiting age-matched pediatric controls can be challenging and for this reason it is important that age-specific pediatric structural templates and quantitative MRI atlases are available to researchers.

Diffusion tensor imaging (DTI) is frequently used to assess white matter microstructure and pediatric developmental trajectories.¹² DTI analysis involves fitting a Gaussian diffusion tensor to each voxel of diffusion-weighted MRI data to extract a principal diffusion direction and other metrics. DTI has good scan-rescan reproducibility (coefficient of variation 2% in cerebral white matter).¹³ In childhood development studies, fractional anisotropy (FA), a scalar measure of the degree of anisotropic diffusion, is found to rapidly increase during childhood and adolescence while the mean diffusivity (MD) decreases.^{9,14–16} MD is calculated from the trace of the diffusion tensor, incorporating both the axial diffusivity (AD) along the principal diffusion direction, and the radial diffusivity (RD) perpendicular to the principal diffusion direction. FA, MD, AD, and RD have also been linked to a range of cognitive processes including math¹⁷ and reading ability.¹⁸ Although DTI is sensitive to a range of white matter tissue changes, it is not specific to myelin. Diffusion anisotropy is present in white matter tissue in the absence of myelin and DTI metrics reflect many aspects of white matter microstructure, including axon density,

diameter, alignment, and myelination, many of which change during brain maturation.^{19,20} Atlases of DTI metric distributions in healthy adult^{21–24} and pediatric brains (for some age groups) are available.^{16,25,26}

Myelin Water Imaging (MWI) has been shown to have a high specificity to myelin. For this technique, T₂ relaxation data are acquired using a multi-echo Gradient and Spin Echo (GRASE) sequence. The T₂ decay is fit with multiple T₂ components to isolate signal from the short-T₂ water trapped between myelin lipid bilayers.^{27–29} Myelin water fraction (MWF), defined as the fraction of the total signal which has T₂ < 40 milliseconds at 3 Tesla, has been histologically validated as a marker of myelin density in human tissue,^{30–32} and has good scan-rescan reproducibility (coefficient of variation 4% in cerebral white matter).³³ MWI has been used to investigate myelin in a wide range of diseases including multiple sclerosis, neuromyelitis optica spectrum disorder, schizophrenia, Niemann-Pick disease, prenatal alcohol exposure, amyotrophic and primary lateral sclerosis, Krabbe disease, autism, stroke, and traumatic brain injury.^{34–47} Rapid myelination has been demonstrated in early childhood and adolescence using multicomponent driven equilibrium single pulse observation of T₁ and T₂ (mcDESPOT), an alternate approach to MWI using a steady-state acquisition.^{48–53} Adult multi-echo T₂-based MWF atlases in both brain and cervical spinal cord have recently been

Table 1. ROI Values Given as Mean $\pm$ Standard Deviation (Coefficient of Variation [%]) for the Adult (Number of Participants = 50) and Pediatric (Number of Participants = 20) Atlases, using Custom ROIs Restricted to the Brain Region Covered by the MWI Sequence

White Matter ROI	PEDIATRIC ROI volume (mm ³)	ADULT ROI volume (mm ³)	ADULT MWF (%)	PEDIATRIC MWF (%)	p-value (t-test for adult/pediatric comparison)	PEDIATRIC FA	PEDIATRIC MD (10 ⁻³ mm ² /s)	PEDIATRIC AD (10 ⁻³ mm ² /s)	PEDIATRIC RD (10 ⁻³ mm ² /s)
Genu of the corpus callosum	817	1158	6.0 $\pm$ 3.1 (52%)	4.7 $\pm$ 4.4 (93%)	.180	.54 $\pm$.07 (13%)	.82 $\pm$.05 (6%)	1.40 $\pm$.12 (9%)	.53 $\pm$.07 (13%)
Body of the corpus callosum	7926	11860	8.8 $\pm$ 3.2 (36%)	5.7 $\pm$ 3.5 (62%)	.001*	.58 $\pm$.07 (12%)	.84 $\pm$.07 (8%)	1.48 $\pm$.13 (9%)	.52 $\pm$.08 (16%)
Splenium of the corpus callosum	3596	5269	10.2 $\pm$ 2.9 (28%)	6.6 $\pm$ 3.3 (50%)	<.001*	.65 $\pm$.07 (11%)	.80 $\pm$.07 (9%)	1.51 $\pm$.14 (9%)	.45 $\pm$.09 (20%)
Anterior Corona Radiata	2779	3989	6.9 $\pm$ 2.6 (38%)	4.8 $\pm$ 1.6 (33%)	.001*	.37 $\pm$.06 (17%)	.77 $\pm$.04 (5%)	1.08 $\pm$.08 (8%)	.61 $\pm$.05 (9%)
Superior Corona Radiata	9209	14846	12.1 $\pm$ 2.9 (24%)	8.6 $\pm$ 1.9 (22%)	<.001*	.46 $\pm$.06 (14%)	.72 $\pm$.03 (5%)	1.11 $\pm$.08 (7%)	.53 $\pm$.05 (9%)
Posterior Corona Radiata	4740	7182	9.6 $\pm$ 2.5 (26%)	6.5 $\pm$ 2.1 (32%)	<.001*	.44 $\pm$.08 (17%)	.79 $\pm$.06 (7%)	1.18 $\pm$.10 (9%)	.59 $\pm$.07 (12%)
Cingulum	2937	3446	5.3 $\pm$ 2.7 (51%)	4.3 $\pm$ 2.9 (68%)	.187	.44 $\pm$.08 (18%)	.78 $\pm$.06 (7%)	1.17 $\pm$.11 (10%)	.58 $\pm$.07 (13%)
Superior Longitudinal Fasciculus	7649	10864	11.8 $\pm$ 2.8 (24%)	7.5 $\pm$ 1.7 (23%)	<.001*	.43 $\pm$.07 (17%)	.73 $\pm$.05 (6%)	1.08 $\pm$.09 (9%)	.56 $\pm$.06 (12%)
Corticospinal Tracts	1059	1785	16.2 $\pm$ 2.8 (17%)	11.7 $\pm$ 2.1 (18%)	<.001*	.52 $\pm$.07 (13%)	.72 $\pm$.03 (4%)	1.18 $\pm$.09 (7%)	.49 $\pm$.05 (9%)

*Indicates statistical significance at the $p < .05$ level.

published,^{54,55} but, to our knowledge, no such atlases exist for children.

The goal of our study was to create pediatric DTI and MWF atlases for healthy children, along with a corresponding structural template, to use as normative data for registration and reference in pediatric brain research. As an initial study, we chose a homogeneous cohort in the age range of 9-10 years old. This age group is of interest as it has never been studied with MWI before and is a period of rapid learning and cognitive maturation before adolescence. The structural, DTI, and MWF atlases were produced using Advanced Normalization Tools (ANTs)^{56,57} software along with FSL affine registration.^{58,59} ANTs outperforms cross-sectional template analysis methods, especially with pediatric brain data.⁶⁰⁻⁶² ANTs transformations allow for large deformations, maximizing anatomical alignment while preserving natural variations within a population. Our DTI and MWI atlases can be used to produce Z-score maps for individual patients to enable patient monitoring.

Methods

Participants and Data Acquisition

Twenty healthy children (mean age: 9.6 years, range: 9.2-10.3 years, 4 females) in Grade 4 were imaged using a 3.0 Tesla MRI scanner (Philips Achieva 3T, Best, The Netherlands) with an eight channel head coil. Ethical approval was granted by the Clinical Research Ethics Board at the University of British Columbia. Written, informed consent for the participation in this study was obtained from the legal guardians of the children. Imaging data included:

- (1) 3D-T₁ (IR-TFE, TR/TE = 9.816/5.70 milliseconds, shot interval = 3,000 milliseconds, TI = 876.6 milliseconds, acquired resolution = 1.1 $\times$ 1.1 $\times$ 1.1 mm³, 136 slices, whole brain coverage).

- (2) MWI (48-echo 3D GRASE, TR/ Δ TE = 1,000/8 milliseconds, acquired resolution = 1.5 $\times$ 1.8 $\times$ 4.4 mm³, reconstructed resolution = 1.1 $\times$ 1.1 $\times$ 4.4 mm³, 10 slices with the inferior border of the volume along the inferior edge of the genu/splenium of the corpus callosum, covering a 211 $\times$ 181 $\times$ 44 mm³ field of view which was 38-47% of the total brain volume).
- (3) DTI (diffusion-weighted 2D SE-EPI, TR/TE = 3,600/68 milliseconds, b -values = 0 (acquired at start) and 1,000 seconds/mm², 15 diffusion directions, acquired resolution = 2.2 $\times$ 2.2 $\times$ 2.2 mm³, reconstructed resolution = 1.1 $\times$ 1.1 $\times$ 2.2 mm³, 60 slices, whole brain coverage).

MWI and DTI Analysis

MWI data were corrected for stimulated echo artifacts using an extended phase graph algorithm to estimate the refocusing flip angle in each image voxel.⁶³ Data were then fit with a multiexponential decay using a non-negative least squares (NNLS) algorithm (in-house MatLab software which can be requested from <https://mrresearch.med.ubc.ca/news-projects/myelin-water-fraction/>) for 40 T₂ components logarithmically spaced between 15 milliseconds and 2 seconds.²⁸ Maps of MWF, defined as the signal fraction with T₂ < 40 milliseconds, were produced for each participant. These maps then underwent a novel non-local spatial regularization to produce a smoother, more globally consistent metric map.⁶⁴ DTI data were motion and eddy-current corrected with FSL's "eddy" tool and Gaussian diffusion tensors were fit to the data in each voxel using FSL's "diftit" which were used to produce FA, MD, RD, and AD maps for each participant.^{65,66}

Atlas Creation

All 20 3D-T₁ structural images were corrected for low-frequency bias and normalized to their mean intensity before being combined to create a representative pediatric template

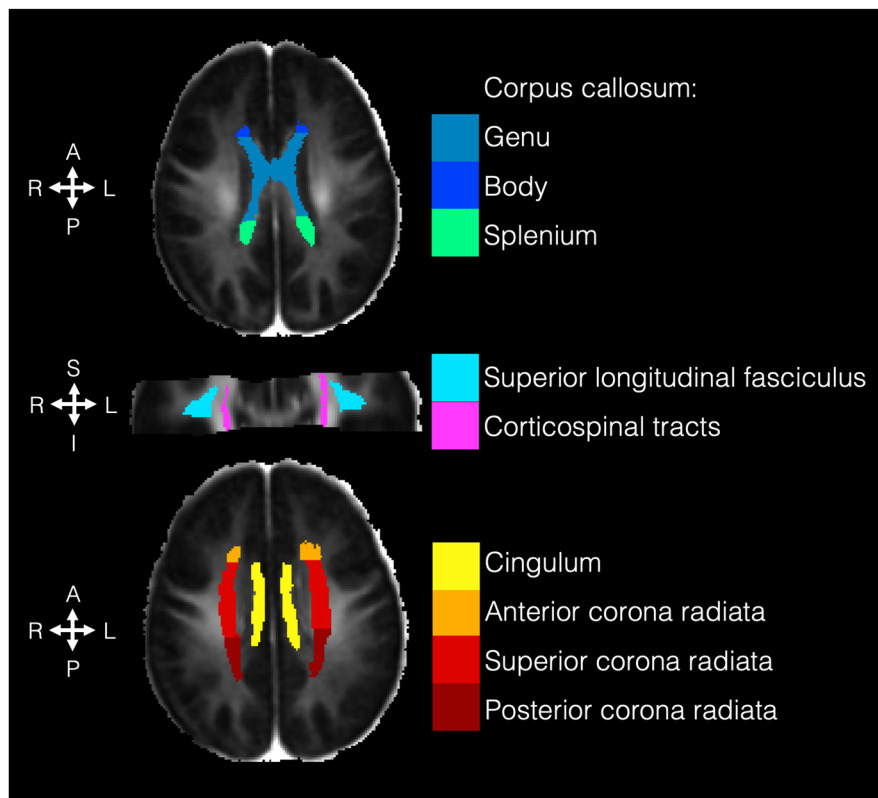


Fig 2. The three commissural and six bilateral regions of interest (ROIs) chosen for our study overlaid on the pediatric myelin water imaging data. The superior-inferior extent of the myelin water imaging acquisition region can be seen. These ROIs were used for the pediatric MWI and DTI analysis and then warped to the adult template space and used to evaluate the adult MWI.

using ANTs.⁶⁷ Template creation involved iterative rigid, affine, and symmetric diffeomorphic normalization transformation. The first echo (echo 1) of the MWI acquisition and FA map for each subject were registered to the corresponding 3D-T₁ scan using an FSL affine registration with 12 degrees of freedom.^{58,59} The affine transformation created for the FA map was used to align the RD, MD, and AD maps and the affine transformation created for echo 1 was used to align the MWF map, to each participant's 3D-T₁ scans. All of the metric maps were then transformed to the template space using the ANTs transformations previously produced (Figure 1 shows this process). Finally, voxel-wise metric mean and standard deviation maps were calculated, combining the data from all twenty subjects.

ROI and Statistical Analysis

Nine white matter ROIs which were covered by the MWI acquisition region, three commissural and six bilateral, were chosen from the JHU White Matter Atlases (Table 1).^{68–70} The commissural corpus callosum ROIs (genu, body, and splenium) extended laterally in the left-right direction. The JHU template was registered to pediatric structural template using FSL non-linear registration.^{71–73} The non-linear warp produced was then used to transform all nine ROIs to the pediatric template space where they could be applied to the metric maps. The structural template was segmented using FSL's FAST algorithm.⁷⁴ The white matter mask produced was thresholded at a 90% white matter probability and used to mask the ROIs to include only white matter voxels. The ROIs were further masked by the re-

gion of the brain covered by the MWI acquisition which fully encompassed the body of the corpus callosum, and the superior and posterior corona radiata. All other ROIs had some voxels outside of the MWI acquisition region. Final ROIs used on both the adult and pediatric MWI and DTI atlas analysis can be seen in Figure 2. Average mean and standard deviation of MWF, FA, MD, RD, and AD in each ROI was calculated from the atlases (Table 1). The same ROIs which had been applied to the pediatric maps were used to analyze adult MWF from an atlas created using a 32-echo 3D GRASE acquisition protocol collected on the same scanner from 50 healthy adults (mean age: 25 years, range: 17–42 years, 25 females).⁵⁴ Two-tailed paired *t*-tests were performed to compare pediatric MWF with adult MWF. DTI data were not available from this adult sample and so pediatric DTI metrics were not compared to adult values. Spearman correlations between pediatric MWF and DTI metrics and also between adult and pediatric MWF were examined.

Results

Atlases

The pediatric structural template and MWF and DTI atlases are publicly available as NIFTI files at <https://sourceforge.net/projects/pediatric-mri-myelin-diffusion/>. As can be seen qualitatively in Figure 3, the grey matter/white matter delineation shown by the MWI and DTI atlases were anatomically similar, however, MWF appeared more homogeneous throughout

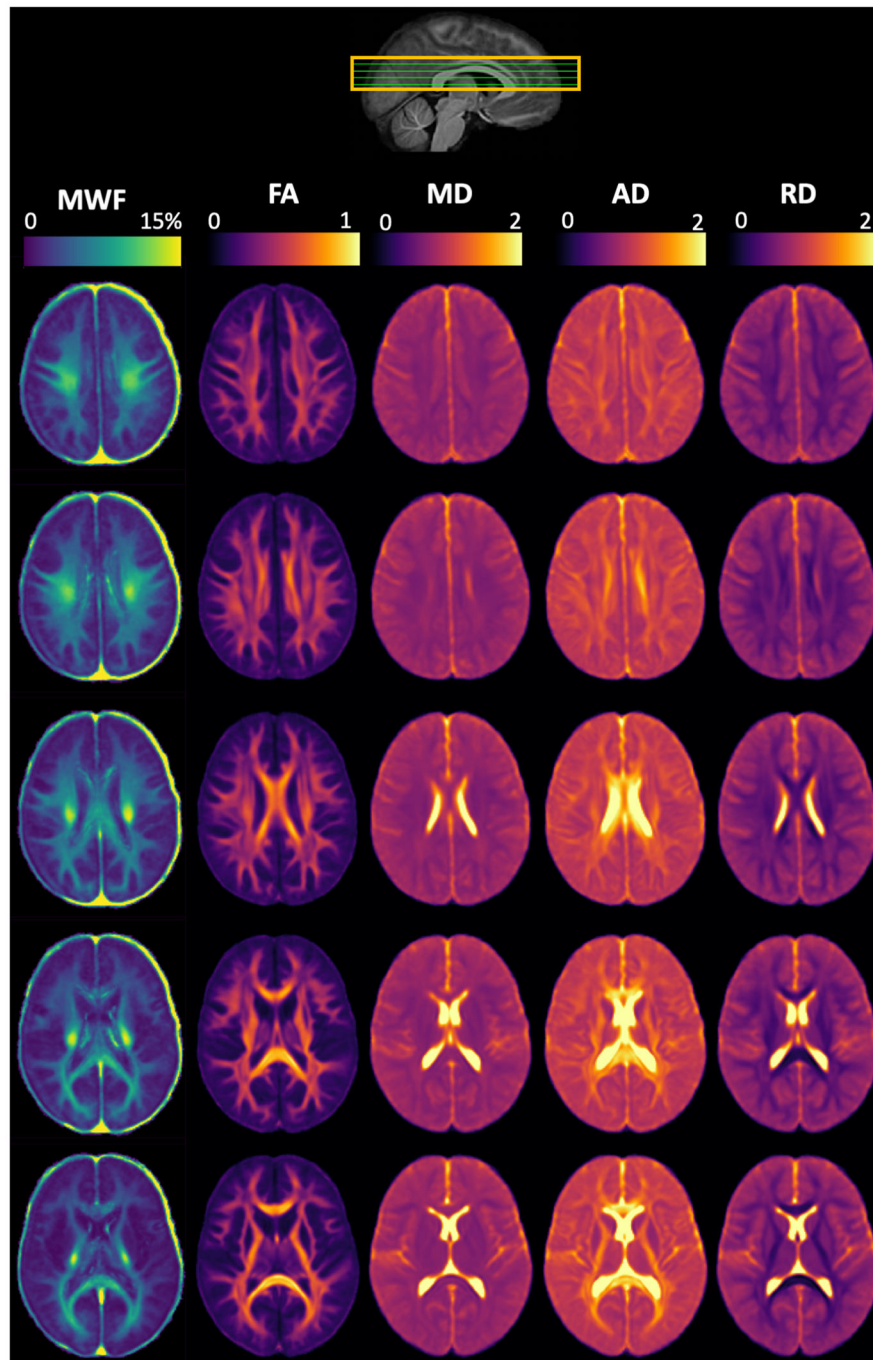


Fig 3. Five representative axial slices of the Myelin Water Fraction (MWF), Fractional Anisotropy (FA), Mean Diffusivity (MD), Axial Diffusivity (AD), and Radial Diffusivity (RD) atlas metric maps. Yellow box shows superior/inferior coverage of the Myelin Water Imaging acquisition. MD, AD, and RD are in units of $10^{-3} \text{ mm}^2/\text{s}$.

white matter than the DTI metrics, with the exception of very high MWF in the corticospinal tracts. FA appeared much higher in the genu, body, and splenium of the corpus callosum and lower in the anterior and posterior corona radiata. ROI metric values (Table 1) confirmed visual observations, with MWF being highest in the corticospinal tracts, superior corona radiata, and superior longitudinal fasciculus while FA was highest in the genu, body, and splenium of the corpus callosum.

Standard deviation in all metric maps was low throughout the brain (less than 4% MWF, .08 FA, $.07 \times 10^{-3}$

$\text{mm}^2/\text{seconds}$ MD, $.14 \times 10^{-3} \text{ mm}^2/\text{seconds}$ AD, and $.09 \times 10^{-3} \text{ mm}^2/\text{seconds}$ RD) and without clear anatomical structure, indicative of good anatomical alignment (representative slices are shown in Figure 4). DTI metrics were more consistent across participants with FA, MD, RD, and AD having a smaller mean voxel-wise standard deviation than MWF.

When the bilateral ROIs were analyzed separately, two-tailed paired *t*-tests found no significant differences between left and right regions for any of the ROIs in MWF, FA, MD, AD, or RD.

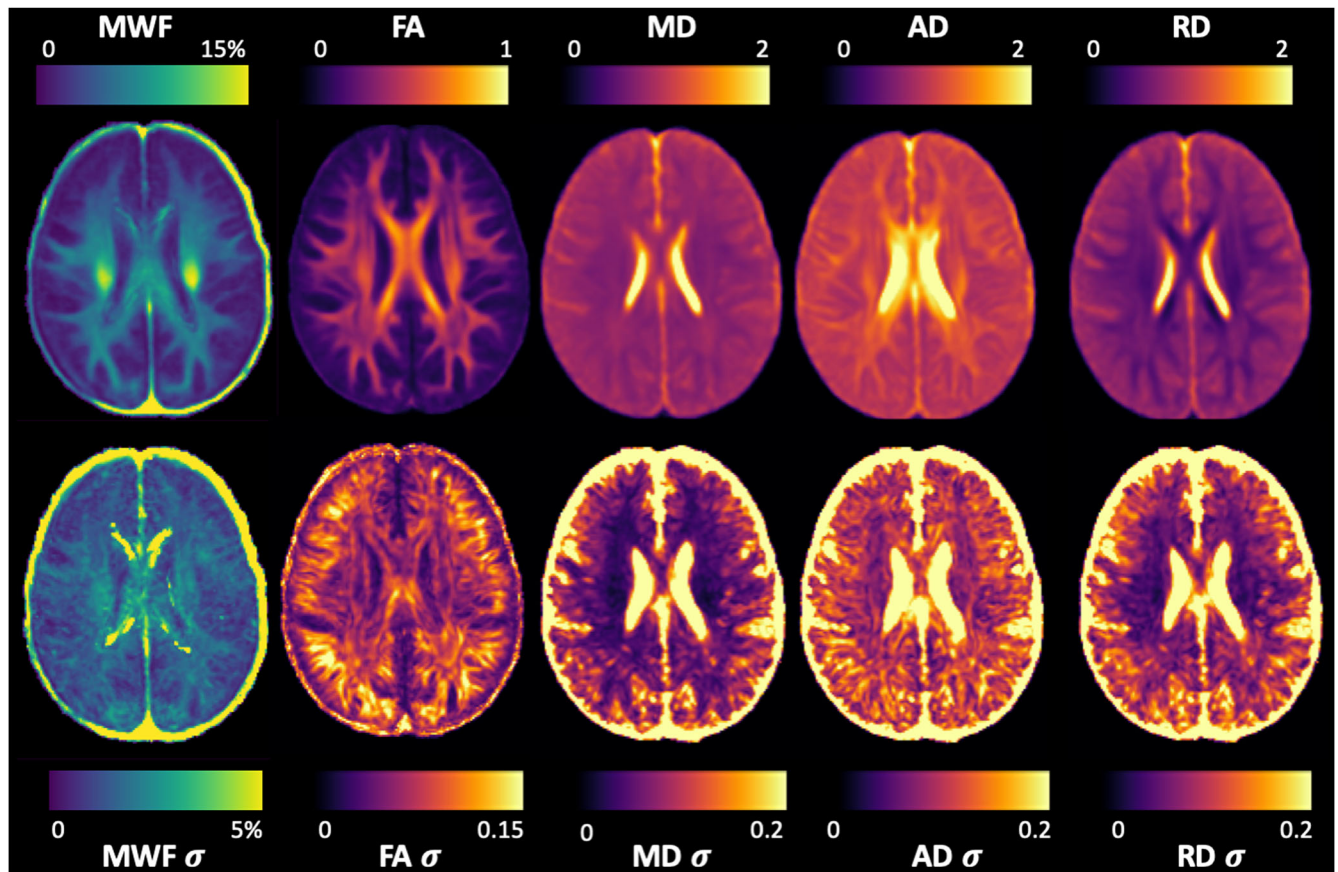


Fig 4. Single slices displaying Myelin Water Fraction (MWF), Fractional Anisotropy (FA), Mean Diffusivity (MD), Axial Diffusivity (AD), and Radial Diffusivity (RD) mean (top) and standard deviation σ (bottom). MD, AD, and RD are in units of $10^{-3} \text{ mm}^2/\text{s}$.

Pediatric and Adult MWF Comparison

Adult ROI MWF values were between 23 and 58% higher than pediatric MWF, with the greatest difference in the body and splenium of the corpus callosum and the superior longitudinal fasciculus. This difference was significant in seven of the nine ROIs investigated (Figure 5) but not in the cingulum and the genu of the corpus callosum. There was a very strong correlation between the pediatric and adult MWF values ($R^2 = .957$, $P < .0001$).

MWF and DTI Metric Comparison

At the $\alpha = .05$ significance level, only MD ($R^2 = .486$, $P = .037$) was significantly correlated with MWF in the nine white matter ROIs investigated (Figure 6), while FA, AD, and RD were not.

Discussion

To our knowledge, this is the first MWF and DTI metric atlas, and corresponding structural template, for children aged 9-10 years old. MWF had a slightly higher standard deviation than the DTI metrics. This may be caused by the small myelin signal resulting in a lower signal to noise ratio in the MWI acquisition. Alternatively this may be reflecting a larger natural population variation in MWF. Standard deviation in all maps was higher in the cortical areas (bright rim in Figure 4) caused by larger registration error near the edge of the brain which occurs due to the natural variation of gyri and sulci.

We compared our atlas with the only other published MWI data from children in this age range. Geeraert et al. studied a pediatric cohort with a slightly higher mean age (11.5 years) and a larger range (8-13 years) using a mcDESPOT acquisition on a 3T GE Discovery system.⁵³ The mcDESPOT-derived myelin volume fraction (VF_m) values were consistently much higher (30-250%) than the T_2 -based MWF values, as has been previously reported.⁷⁵ We found an increase in MWF moving from anterior to posterior of the corpus callosum, matching the variation in MWF measured in adults.⁵⁴ In contrast, Geeraert et al. found that the genu had the highest VF_m . When comparing VF_m and T_2 -based MWF, the effect of exchange may also play a role. It has been shown that smaller axon diameters and thinner myelin sheath results in more intercompartmental proton exchange and, therefore, smaller MWF so this may be contributing to our results.⁷⁶ On the other hand, mcDESPOT-derived MWF may be affected by magnetization transfer effects due to the use of short, high amplitude radiofrequency pulses which may be contributing to the much higher MWF values reported.⁷⁷ mcDESPOT fitting has also been found to have an inherent and unpredictable bias which depends on initial search bounds.⁷⁸

As in previous adult studies, high MWF values were observed in the corticospinal tracts. This may be due to the intraextracellular water signal in this region which has been shown to have a broader T_2 range and is frequently split into two separate peaks, suggesting that intra- and extracellular environments have more distinct T_2 times in the corticospinal

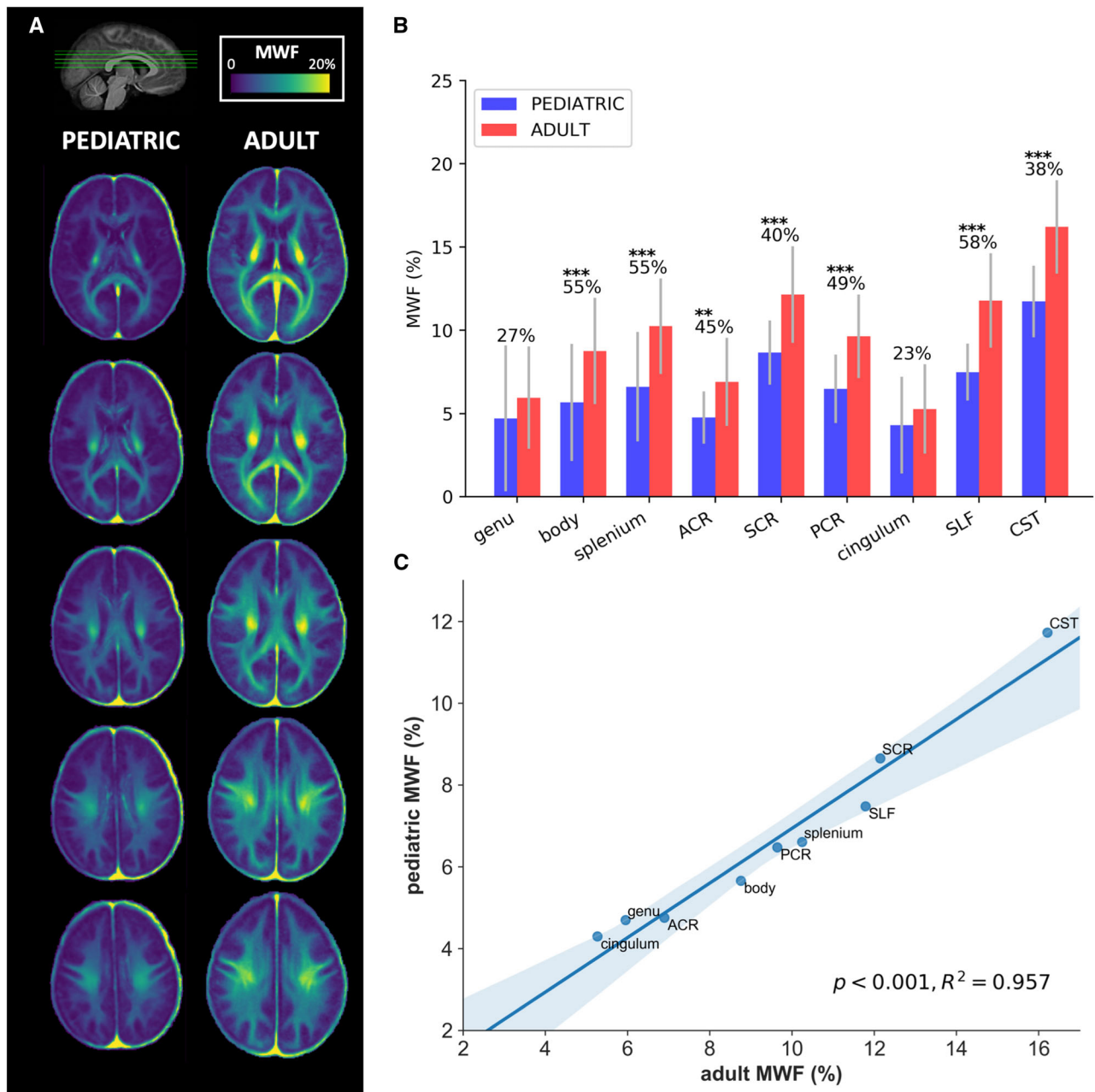


Fig 5. (A) Adult and pediatric Myelin Water Fraction (MWF) slices for qualitative comparison. (B) Pediatric MWF (number of participants = 20, 9-10 years) was significantly lower than adult MWF (number of participants = 50, 17-42 years) in seven out of nine of the ROIs investigated ($***P < .001$, $**P < .01$). Percentage increases, as a fraction of the pediatric MWF, are shown above the bars, ranging from 23 to 58% higher in the adult brain. (C) There was a very strong correlation between adult and pediatric MWF. ACR = anterior corona radiate; CST = corticospinal tract; PCR = posterior corona radiate; SCR = superior corona radiate; SLF = superior longitudinal fasciculus.

tracts. This broad peak can extend below the $T_2 = 40$ millisecond myelin water threshold and so interpretations of this ROI result should be made with caution.⁷⁹

Myelin in Children versus Adults

MWF in children aged 9-10 years was significantly lower than in adults in most of the ROIs investigated which was expected, because cognitive development and brain maturation continue throughout the teenage years.^{2,8,9} Similar myelin increases during development are also seen using the mcDESPOT approach to MWI.^{49,50,52}

An interesting result is the very strong correlation between adult and pediatric MWF across the nine white matter ROIs. During earlier development, histological studies have shown a clear spatiotemporal pattern of myelination over time. Myelination begins in the cerebellum and brain stem before birth and progresses through the brain from posterior to anterior and from the center out.⁸⁰ However, this strong correlation between MWF in adults and 9-10 years old children suggests that, by this stage of development, the white matter ROIs maintain their relative myelination hierarchy into adulthood.

There were some differences between the acquisition and processing protocols used to create the adult and pediatric

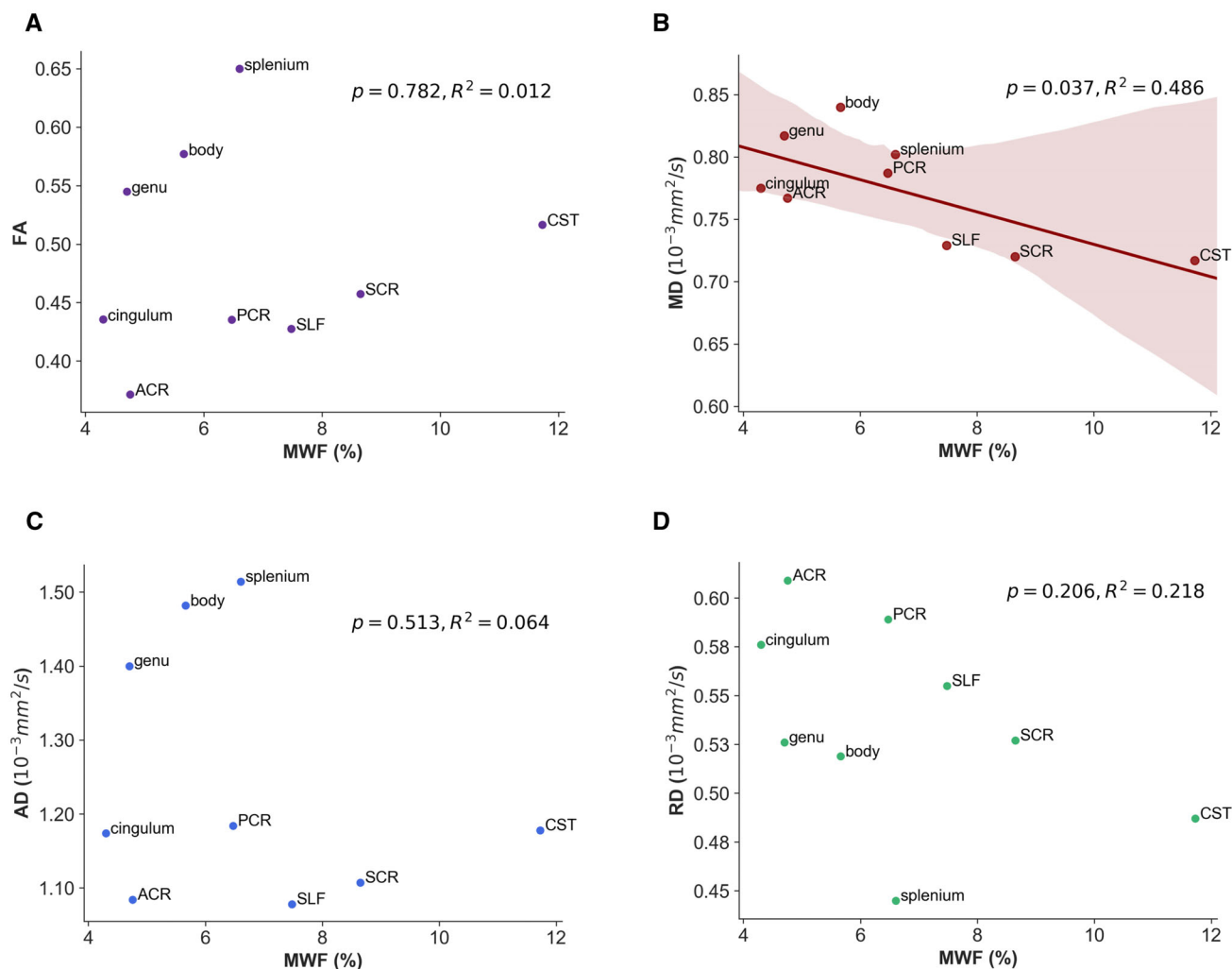


Fig 6. Significant correlation between Mean Diffusivity (MD) (B) and Myelin Water Fraction (MWF). No significant correlation between Fractional Anisotropy (FA) (A), Axial Diffusivity (AD) (C), or Radial Diffusivity (RD) (D) and MWF. The regions of the corpus callosum (genu, body, and splenium) do not follow the trend shown by the other ROIs, as they have high diffusion anisotropy but relatively low MWF. ACR = anterior corona radiate; CST = corticospinal tract; PCR = posterior corona radiate; SCR = superior corona radiate; SLF = superior longitudinal fasciculus.

atlases. Data for both studies were acquired on the same Philips Achieva 3T scanner, however, the adult MWF atlas was created using 32-echo GRASE data with 10 millisecond spacing, whereas the pediatric atlas was created using 48-echo GRASE data with 8 millisecond spacing. These two echo-acquisition schemes have been contrasted and found to be comparable.⁸¹ The adult data were acquired at $.99 \times .99 \times 5 \text{ mm}^3$ resolution, reconstructed to $1 \times 1 \times 2.5 \text{ mm}^3$ and so is slightly higher resolution. This should not have a large impact on MWF maps, although smaller voxel sizes will reduce partial volume effects. The range of T_2 components used for fitting (15 milliseconds–2 seconds) and the MWF T_2 range (15–40 milliseconds) were the same for both analyses but the pediatric atlas included a novel non-local spatial regularization which is thought to give more globally consistent MWF values while preserving fine structure detail.⁶⁴ This should not affect the ROI values found. Finally, it is worth noting that the adult MWF ROI values reported here are different from those reported by Liu et al. although both studies use the same adult MWF brain atlas. This discrepancy stems from the difference in ROIs used; our ROIs were limited

by the pediatric MWI acquisition region so in some cases did not encompass the entire white matter structure. We used the same ROIs for the adult and pediatric data which enabled a direct comparison between the two groups.

Complementary Information from MWF and DTI Metrics

MD was the only DTI metric found to be significantly correlated with MWF when analyzing all nine ROIs. If a Bonferroni correction for multiple MWF/DTI metric comparisons was performed, reducing the significance threshold to $P < .05/4 = .0125$, this correlation became nonsignificant. These results show that DTI and MWI are affected by different aspects of the tissue microstructure and so should be used in conjunction to characterize tissue.

The three ROIs of the corpus callosum lay outside the linear relationship shown by the other ROIs for MWF versus FA, AD, and RD (Figure 6). The corpus callosum had significantly higher FA than expected given its fairly low MWF (5–7%). This finding, which agrees with Mädlér et al.'s comparison in adults, is thought to be due to tightly-packed, coherently-oriented axons

in this region which increase the diffusion anisotropy independently of the myelination.⁸²

Limitations

There are several limitations which should be considered when interpreting our study. (1) Although reasonably sized for a pediatric study, our atlas has a fairly small sample size which may make it less reliable for representing the whole population of the 9-10 years old age group. (2) Our MWI sequence did not provide whole brain coverage. The 10 slices of MWI data collected encompassed 38-47% of the total brain volume and contained many important white matter structures. However, some of our ROIs did not encompass the full white matter structure, and there were brain regions beyond our field of view which could not be investigated. More recent versions of the GRASE MWI sequence can provide full-brain coverage with a 3D or 2D sequence in under 10 minutes.^{83,84} (3) This study had mostly male participants (16 out of 20) and so we could not perform meaningful analysis of sex differences. However, Geeraert et al. investigated 17 white and grey matter ROIs and found no significant differences in any of them between male and female children aged 8-13 years.⁵³ This suggests that the unequal sex ratio in our data does not reduce its utility for children of both sexes in the 9-10 years age range. (4) Finally, the narrow age range of 9-10 years and is both a benefit and a limitation. Both DTI and MWF metrics change rapidly during adolescence, and so combining data from a range of ages would make the atlas less accurate for any one particular age group. Combining twenty participants in such a narrow age range allowed the creation of a reliable atlas for 9-10 year olds.

Conclusion

We have created normative myelin and diffusion tensor atlases, with a corresponding structural template, for children aged 9-10 years. The atlas is available online for use as a reference in pediatric neuroimaging studies (<https://sourceforge.net/projects/pediatric-mri-myelin-diffusion/>). We found MWF in nine white matter regions was significantly correlated with MD at the $p < 0.05$ level, but not with FA, AD, and RD. Seven of the nine ROIs had significantly lower MWF in the pediatric group than in adults, highlighting the importance of age-specific control data.

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